

Verification is a cost/accuracy problem.

So price the verdict, not the detector.

Jiawei Li · Inference Verification Gym (ivgym) · new measurements on one NVIDIA H100, Qwen3-1.7B audited with a Qwen3-0.6B proxy

A client renting inference cannot see which model ran. Every defence against that is scored on detection accuracy and priced by a rate — FLOPs per sequence, seconds per prefill. Neither is what a client buys. It buys a VERDICT at a false-accusation budget it can live with, and a verdict's price factors into how many tokens the evidence needs and what a token of that evidence costs. Measured that way the ranking of the defences changes, three of them stop being defences at all, and the engineering that matters moves off the estimator and onto the accounting.

141 μs

of H100 time for one verdict on
int4 quant (1 token, activation_difr)

14 of 35

detector × deviation cells cost
infinity: no budget buys the verdict

1.05×

cheaper per token is all the cheap tier
buys — and it needs 13× more of them

1 • THE UNIT OF ACCOUNT

What a client buys is a verdict

cost of a verdict = $(\delta^*/d')^2 \times c(\text{tier})$

tokens of evidence × price per token

$\delta^* = 3.767$ is the batch separation reaching standardized pAUC 0.90 at $\text{FPR} \leq 0.5\%$ — the repo's protocol, not a new one. Both factors are measured below, on one model, in one run.

141 μs

per token, Tier 1:
recompute M

134 μs

per token, Tier 0:
a cheap proxy

0.7 μs

per token, Tier 0:
no model at all

GPU-synchronised, timed alone with the reference cache dropped. The proxy tier is 1.0× cheaper per token — the whole of its advantage, and on the smaller factor.

3 • CHEAPER TOKENS

The same accuracy, 16× cheaper

Budget in prefill tokens, not token counts: reading M at generated position j needs a prefill of everything before it, so depth is cheap and breadth is expensive. No statistics in it — it replicates across deviations by construction.

5 • FEWER TOKENS

Stop early — where the pool can prove it

1.3×

median token saving,
matched at the same power

3 of 22

cells this pool can
resolve at all

173.2k

honest tokens the next
cell would need

Same evidence, stopped when it is enough: 1.13–1.98× fewer tokens, against a fixed design bisected to the same 90% power rather than against $b^*(d')$. That matters, because $b^*(d')$ does not deliver its own spec — inside the ceiling it realizes 0.05–1.00 power and 0.0045–0.0145 false alarms against 0.90 and 0.005. The other 19 cells are not reported: their batches run to 16387% of the pool, and a bootstrap that large measures the pool.

7 • TAKEAWAYS

What we would deploy

- **Price the verdict, not the detector.**
14 of 35 cells cost infinity; the cheapest is 141 μs, on activation_difr — which pays for it in access, not compute.
- **Buy the exponent, not the constant.**
Tokens go as $1/d'^2$, price per token as model size. Effect size wins.
- **Spend on the accounting, not the estimator.**
Prefix scheduling is 16×; three better per-token signals all lost to the plain mean.
- **Stop early — and size the pool to say so.**
Sequential testing saves 1.3× where the pool can resolve it, which is 3 of 22 cells.
- **Measure your own benign floor first.**
It is the price ceiling. Calibrate in the shape you audit in.

2 • THE PLANE

Cost and accuracy do not trade off along a curve

Each mark is one deviation scored by one detector. The axes are the two measured factors; the product is what it costs.

Tier 1 • recompute M

Tier 0 • proxy

Tier 0 • no model

shape = deviation · hollow = unreachable at any budget

Iso-cost diagonals: every point on one costs the same. Measured price takes only 3 values — a tier runs a model or it does not — and the two that do price within 1.05× of each other at batch 1. They are one column, so the vertical axis decides almost everything; marks are nudged sideways only where they would overlap. On the rail: 13 cells at $d' \leq 0$, plus 1 whose d' does not clear its own detector's honest-vs-honest control.

4 • THE CHEAP TIER

A cheaper token is not a cheaper verdict

Tokens the same deviation needs under a full recompute of M and under a client-side proxy that never runs it.

It is not a weak detector — it is a detector of a different thing. On model substitution the same proxy reads AUC 0.998 at 2.3× fewer FLOPs than recomputation, never running M (accept rate 0.776 honest vs 0.700 cheating).

6 • THE CEILING

The accuracy no budget buys

Same weights, same tokens, same noise draw — only the verifier's own forward-pass schedule differs, and every row is honest. Replaying the SAME shape twice returns $d' = 0.000000$ exactly. A floor reaching the orange rule (int4 quant, this run, $d' = 0.0754$; band = its 90% CI) would price a verdict at infinity. None does — the largest is +0.0019. Token-by-token replay shifts the null 45% as far as the attack, but away from it.

Method: one protocol for every number — standardized partial AUC at $\text{FPR} \leq 0.5\%$, threshold from a held-out honest calibration split, winsorized at its 99.9th percentile, batch/pool ratio capped at 10% and enforced in code. Prices are GPU-synchronised wall-clock on an idle H100 at \$2.50/hour; d' comes with a sequence-level bootstrap interval (400 resamples) because tokens inside a sequence are not independent. Pools: $80 \times 256 = 20,480$ tokens per configuration. Reproduce: `python -m experiments.exp_cost_of_a_verdict_gpu · exp_benign_shape_dprime_gpu · exp_sequential_verdict · exp_prefix_cost_gpu` · this poster: `python paper/make_cost_poster.py` · artifacts: `docs/results/{cost_of_a_verdict, benign_shape_dprime, sequential_verdict, prefix_cost_quant2bit, pool_scaling}.json` Limits: attacks are logit-level models of quantisation, not quantised checkpoints; one prompt domain; one accelerator, so the cross-GPU half of the benign floor is still unmeasured — the shapes here are its lower bound. References: Karvonen et al., DiFR (2025) · Leviathan et al. (2022), speculative decoding · Wald (1945), sequential tests · code and every run artifact: the inference-verification (ivgym) repository.